

# SOLAR DEVELOPMENT TOOL

## The Problem

Datacenters are driving up electricity rates across Virginia, and the process of researching solar, comparing products, getting quotes, and understanding costs is tedious and inaccessible. Most existing tools either require technical knowledge or hide results behind email walls and paywalls, creating friction between a homeowner or school and a clear answer about solar potential.

## The Solution

A web-based tool that lets anyone enter a building address, trace their roof on a map, and instantly see estimated energy production, savings, and payback timelines. Long term, the platform organizes community solar investments, allowing schools to host projects and homeowners to subscribe to excess energy

## How It Works

The user enters an address, which is geocoded into coordinates using the Mapbox API. Those coordinates are passed to the NREL PVWatts API alongside roof dimensions drawn directly on the satellite map. PVWatts returns monthly solar production estimates accounting for locaiton, irradiance, and weather. Results are displayed as a monthly production chart and a year-over-year ROI graph.

## What I Built

A functional solar calculator with a map interface, production charts, and ROI projections. The scope grew naturally from a single calculator into a full investment platform vision, with a dashboard designed for admins, schools, and subscribers as the next development phase.

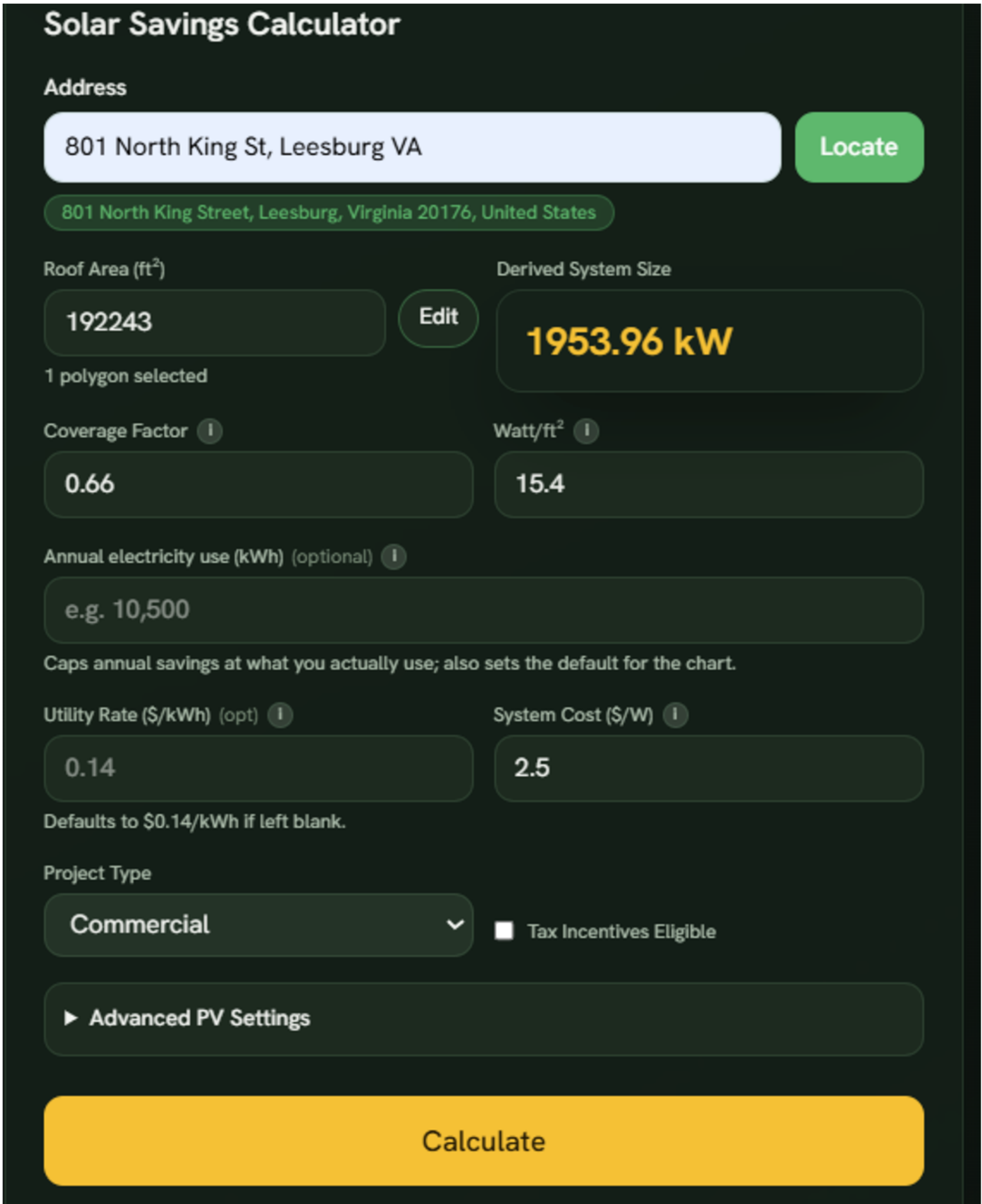


Figure 1. Calculator input interface



Figure 3. Roof area polygon tool

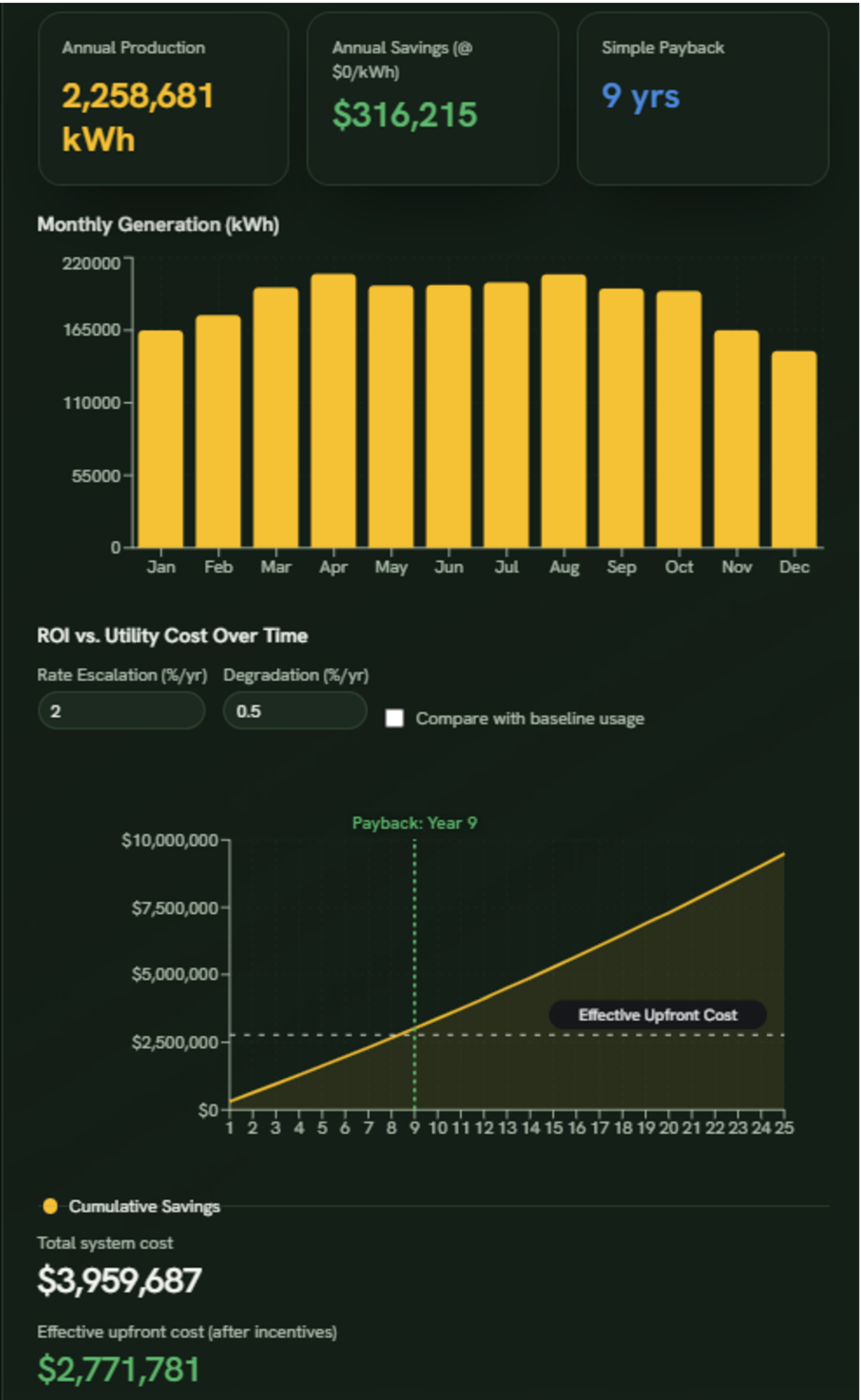


Figure 2. Monthly Production and ROI output charts

## What I Learned

Building firsthand revealed what AI-generated code misses. Catch blocks were swallowing errors silently and there was missed input validation. Reviewing and improving that code taught me more than writing it from scratch would have and in less time. AI is useful for prototyping and scope expansion, but cannot replace the understanding that comes from debugging your own project and building from the ground up.

## What's Next

Integrating SQL databases to persist project and user data. Strengthening the codebase beyond AI scaffolding with proper error handling and logging throughout. Expanding into real commercial deployments, using the calculator as a development tool.

## References

NLR PVWatts (NLR, 2022)  
Mapbox (Mapbox, n.d.)  
JLARC (2024)  
Virginia Department of Energy(n.d.)